Prithvi Chakravarthy

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Education

DePaul University — Chicago, IL
Master of Science in Computer Science — *Sept. 2025 – Present*

- Coursework: Distributed Systems, Applied Algorithms and Structures, DBMS, Real-Time System Architecture

BITS Pilani, WILP — India
Postgraduate Certificate in AI/ML — *Oct. 2023 – Oct. 2024*

- Coursework: Deep Learning, Neural Networks, Text Mining, Feature Engineering, Unsupervised Learning

VIT Bhopal — India
Bachelor of Technology in Computer Science — *Jun. 2018 – Jun. 2022*

- Coursework: Data Structures, Algorithms, DBMS, Operating Systems, Computer Architecture, Soft Computing, Computer Networks, Theory of Computation & Compiler Design

Experience

Designare Solutions Pvt Ltd — Mumbai, India
Full Stack Developer — *March 2022 – August 2025*

- Developed and deployed full stack applications using React, Node.js, Python and PostgreSQL, improving scalability and performance.
- Designed distributed APIs and optimised data pipelines, reducing query time by 25%.
- Managed production deployments on Google Cloud, maintaining 99.9% uptime.
- Implemented secure user authentication and role-based access control (RBAC).
- Collaborated with stakeholders to define requirements and deliver on time.

Projects

Space Invaders Game Engine | *C#, Azul Engine, IrrKlang*

- Architected a C# game engine implementing **14+ GoF design patterns** and **SOLID principles**, utilizing a **State pattern** for scene management and a **Composite tree** for hierarchical game objects.
- Eliminated runtime heap allocation spikes via a universal **Object Pool** system and optimized GPU memory usage through **Proxy and Flyweight** sprite rendering.
- Engineered a decoupled collision system using **Visitor and Observer** patterns for double-dispatch resolution and a **Command pattern** priority queue for timed game events

IQGeo (Telecom/GIS Platform) | *Python, PyTorch, Jupyter*

- Developed backend APIs and React modules to visualize large geospatial datasets, optimizing spatial queries to **reduce data retrieval time by 25%**
- Architected scalable Python (Pyramid) APIs to handle complex spatial queries, improving data retrieval efficiency for telecom clients.

Leisurely | *React, PostgreSQL, GCP*

- Built a custom CMS and high-performance React application, maintaining **99.9% uptime on GCP** while reducing database query latency by 25%
- Integrated Google Analytics to capture user events, leveraging ”what/when” data insights to drive platform engagement and operational efficiency

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, Kotlin, C#
Frameworks/Libraries: React, Node.js, Pyramid, MaterialUI, TailwindCSS, PyTorch, TensorFlow
Databases: PostgreSQL, MySQL, Oracle
Tools: Google Cloud, GitHub, Docker, Jenkins, CI/CD, REST APIs, Jest